- *Multi-modal sensor suite*: Stabilized EO/IR gimbals, SAR/GMTI radars, LiDAR or laser ranging, hyperspectral/multispectral imagers, SIGINT/COMINT/ESM packages, ADS-B/TCAS and sense-and-avoid sensors for airspace integration.
- *Resilient communications hardware*: Multi-band radios (UHF/VHF, L-band), SATCOM terminals (Ku/Ka), 5G/mesh radios for tactical networks, and anti-jamming front ends with frequency agility and directional antennas.
- *Cybersecurity and crypto modules*: Hardware security modules (HSMs), trusted platform modules (TPMs), and quantum-resistant crypto accelerators where required.
- *Launch and recovery systems/support equipment*: Autonomous runway or catapult systems, precision deck/ship recovery interfaces, or parachute/auto-land recovery depending on mission profile.
- *Health sensors and prognostics*: Embedded structural, thermal, and vibration sensing to feed predictive maintenance and in-flight fault response.

5.3.3 Major software and systems components

The software architecture is layered and modular, enabling certification, upgrades, and reuse:

- *Autonomy stack/mission manager*: Implements intent interpretation, mission planning, task allocation (for multi-vehicle teams), and highlevel policy enforcement (ROE, no-strike zones). Often realized as a hierarchy of planners (strategic → tactical → control).
- *Perception and sensor fusion*: ML-based perception pipelines (object detection, tracking, semantic segmentation) fuse EO/IR, SAR, LiDAR,